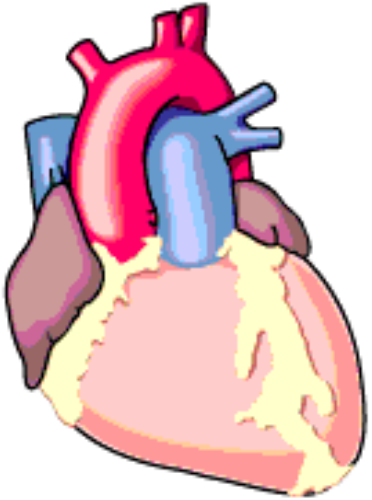
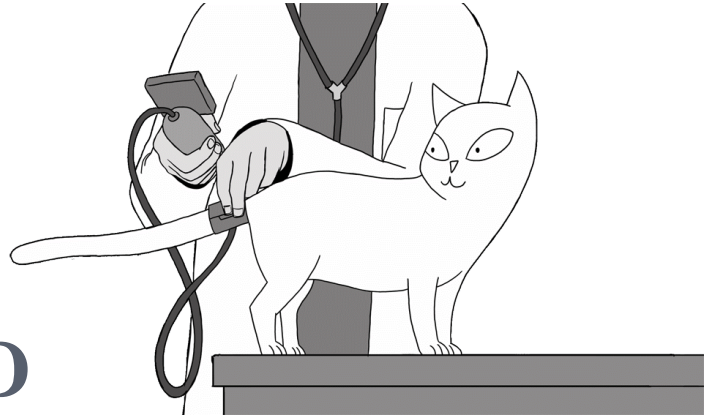
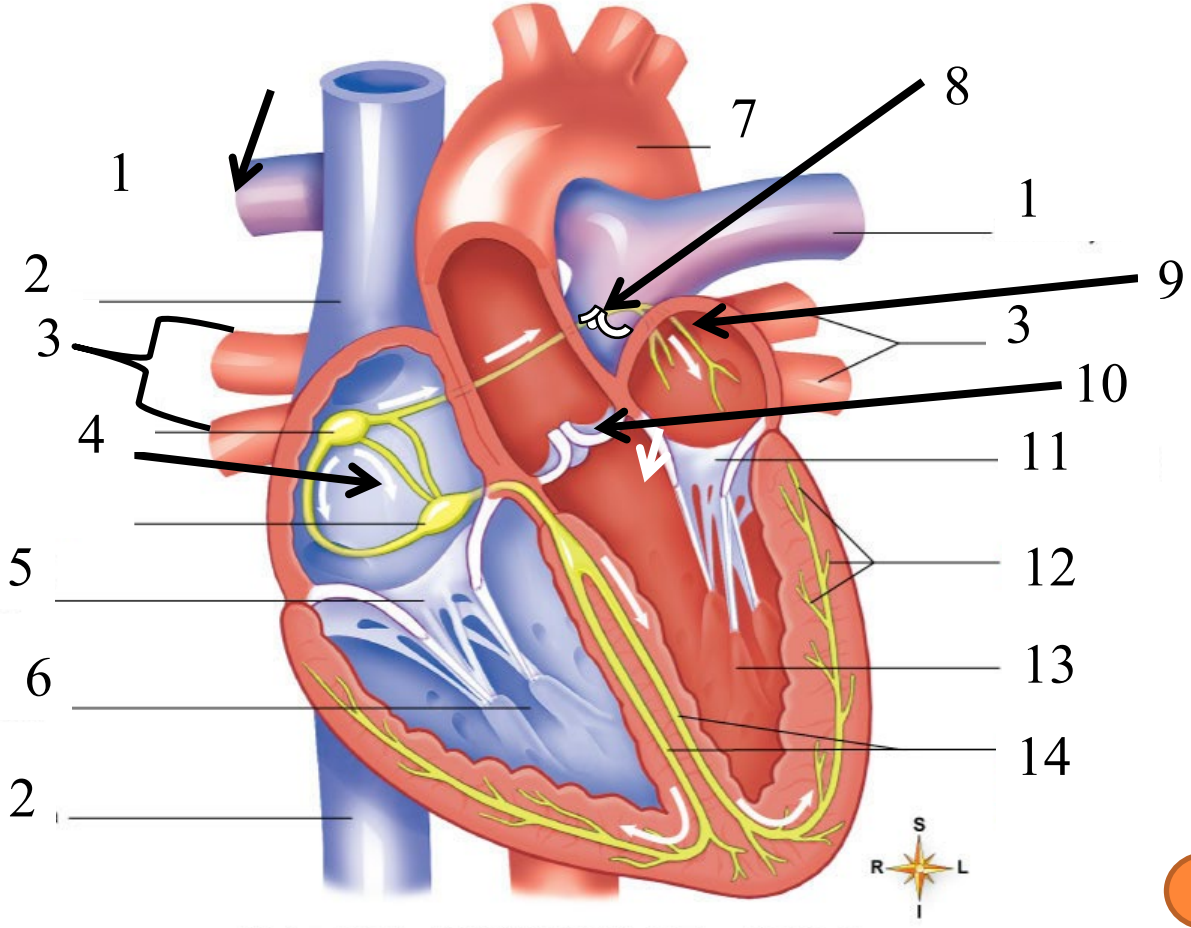




Basic ECG And Blood Pressure

Trace the path of blood



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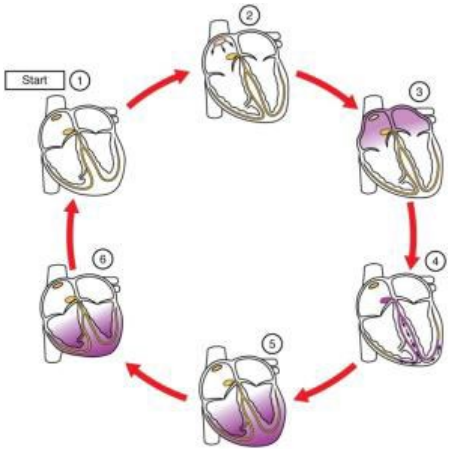
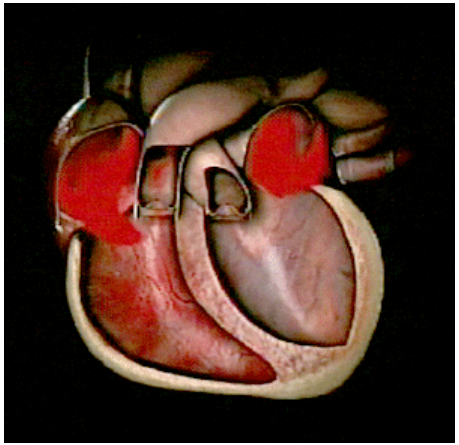
Cardiac Cycle

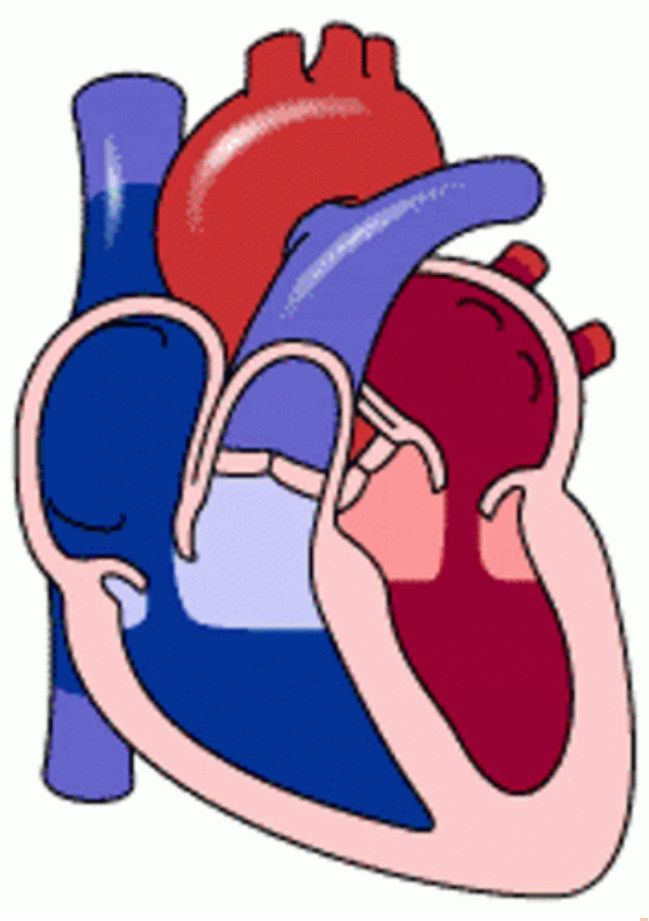
- Coordinated sequence of events that occurs during one heartbeat
  - Contraction and relaxation of the atria and ventricles
  - Movement of the electrical impulses through the conduction system
  - Opening and closing of valves
  - Filling and opening of the heart's chambers





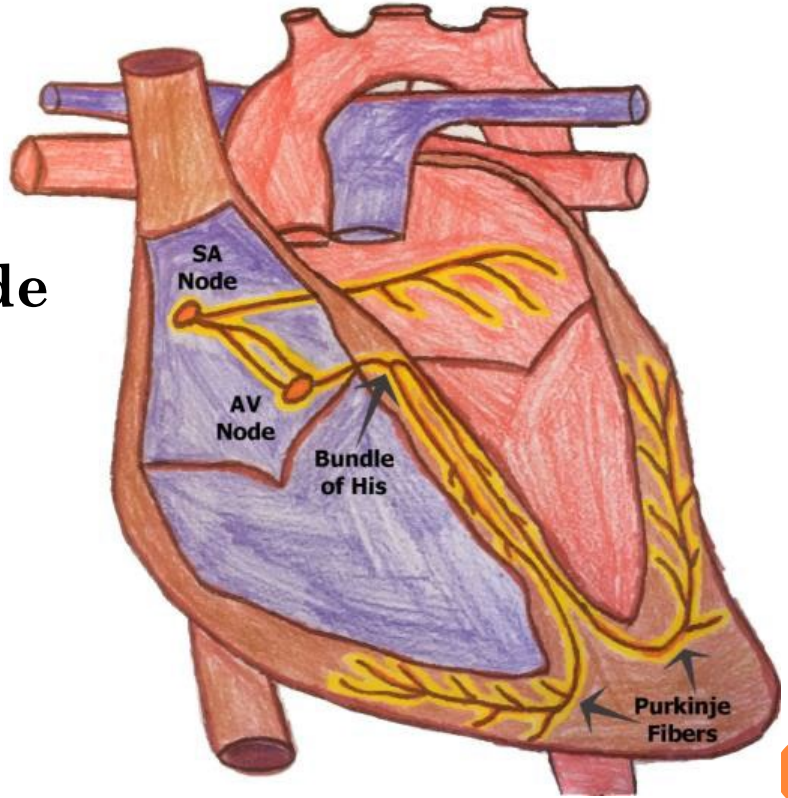
Cardiac auscultation

- Heartbeat sound = “Lub dup”
  - “Lub” = Closure of atrioventricular valves
    - Atria are emptied; blood is loaded into ventricle
  - “Dup” = Closure of aortic and pulmonary valves
    - Blood has exited the ventricles



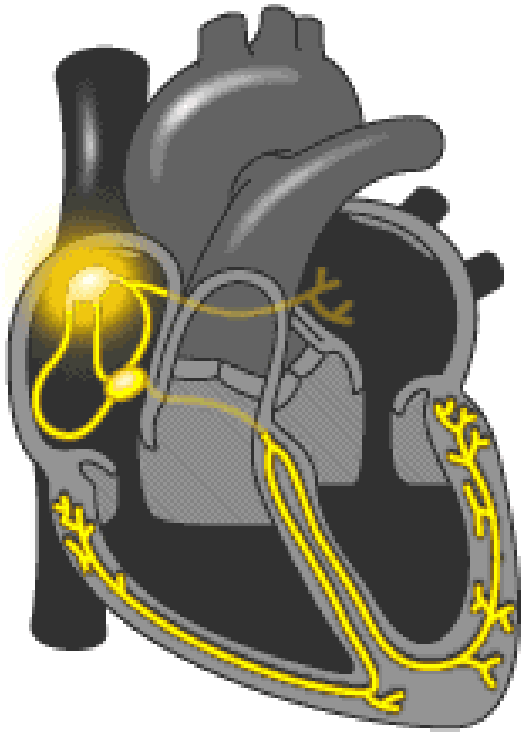
Cardiac Conduction

- Sino Atrial Node (SA)
  - Primary pacemaker
- Atrioventricular (A-V) node
  - Secondary pacemaker
- Bundle of His
- Left and Right Bundle Branches
- Purkinje fibers
  - Tertiary pacemaker



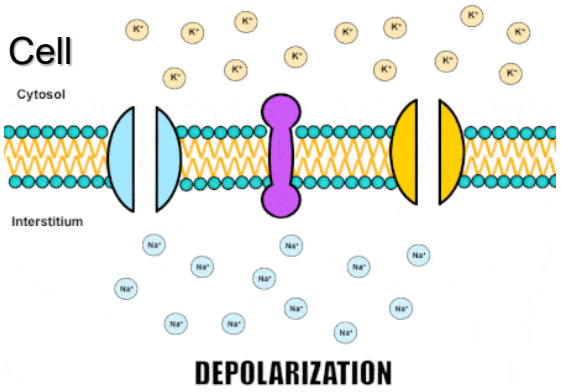
- Depolarization:
  - Release of electrical impulse which cause the cardiac muscle to contract

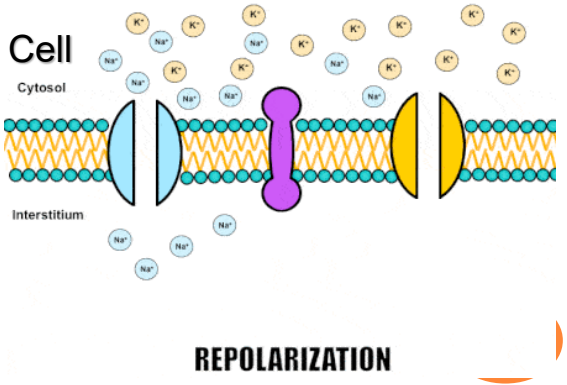
- Repolarization:
  - Recharging of electrical impulse in preparation for depolarization—relaxation of the heart



Depolarization and Repolarization

- Cell at rest more positive ions outside than in
  - Negative inside cell (potassium) and positive outside of cell (sodium)—polarized cell
- Cell receives signal—sodium channel opens
  - Na enters the cell changes polarity—depolarization (+cell, -outside)
- Potassium channel opens
  - Potassium leaves cell changes polarity—repolarization (-cell, +outside)
- Sodium potassium pump
  - Uses ATP puts everything back



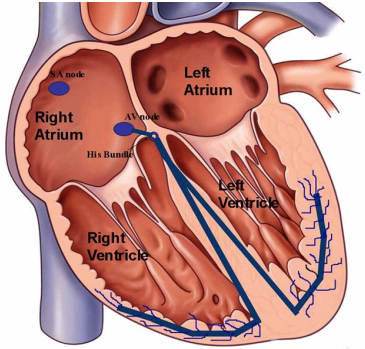


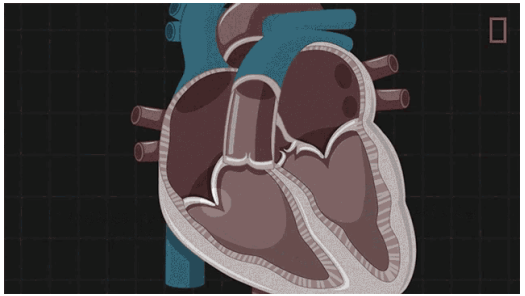
Cardiac Cycle Review

There are a few things that happen in synchronism for every heartbeat to allow for appropriate perfusion to the body.

Mechanical and Electrical

- P wave
  - Atrial depolarization
  - Atrial contraction
- QRS complex—systole—pulse felt
  - Ventricular depolarization
  - Ventricular contraction
  - Atrial repolarization
  - Atrial relaxation
  - AV valves close (lub)
  - Semilunar valves open
- T wave—diastole
  - Ventricular repolarization
  - Ventricular relaxation
  - AV valves open
  - Semilunar valves close (dub)





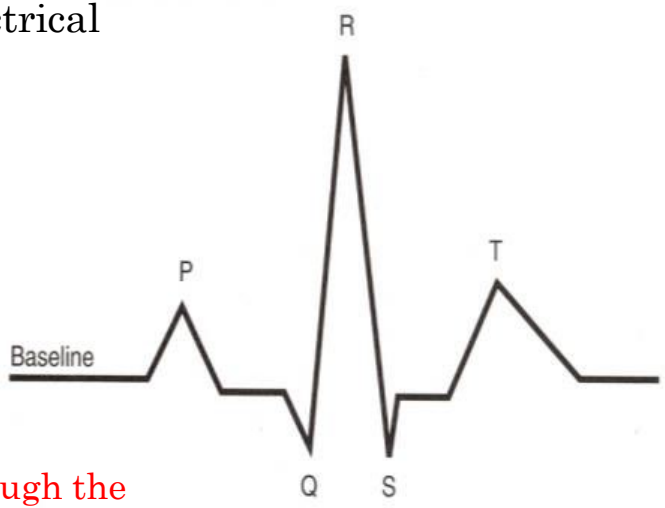
Electrocardiogram (ECG)

- Provides a way to measure the movement of the electrical impulse through the heart during the cardiac cycle

- **P wave**
  - Depolarization of the atria
  - Atrial systole

- **QRS complex**
  - Depolarization of the ventricles
  - Ventricular systole
  - Repolarization of the atria occurs but is not seen through the large wave created by the more muscular ventricles

- **T wave**
  - Repolarization of the ventricles
  - Relaxation/regrouping of the ventricles
  - Ventricular diastole



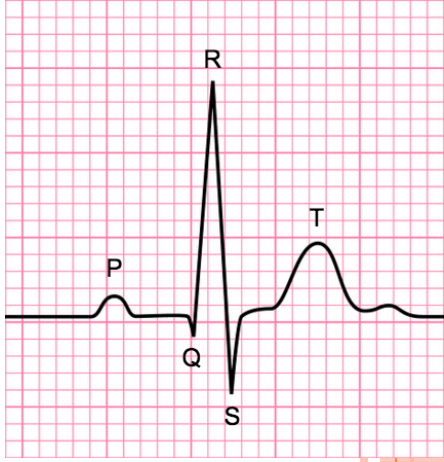
P wave = atrial depolarization

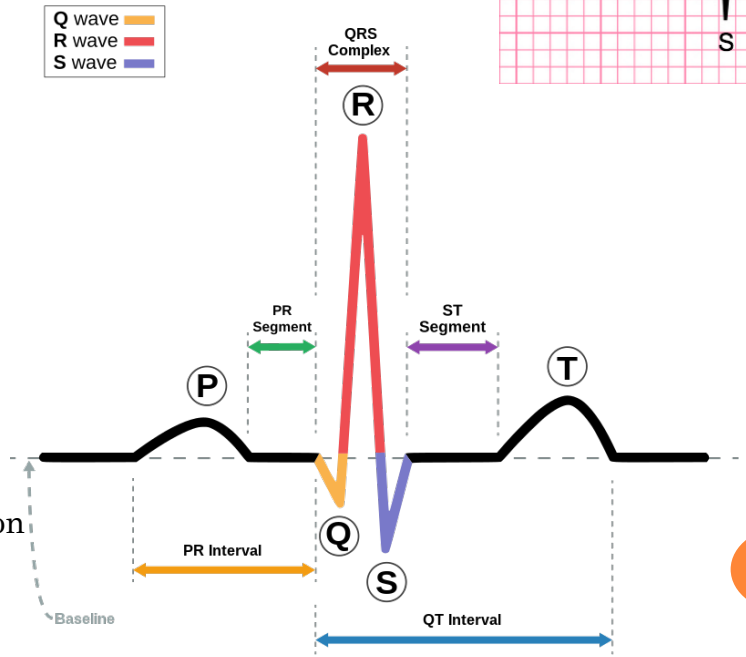
QRS complex = ventricular depolarization

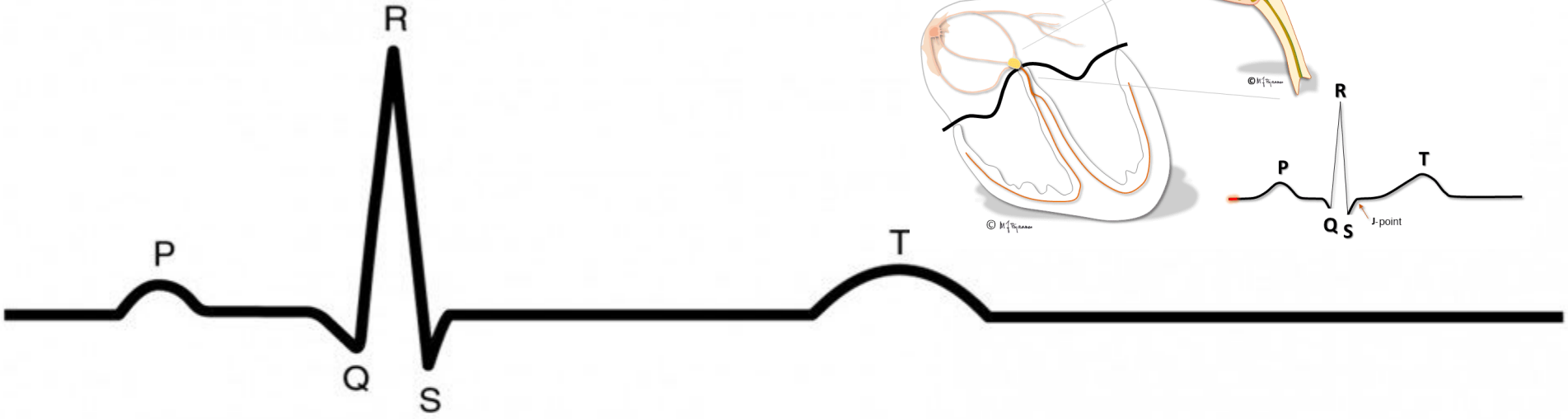
T wave = ventricular repolarization

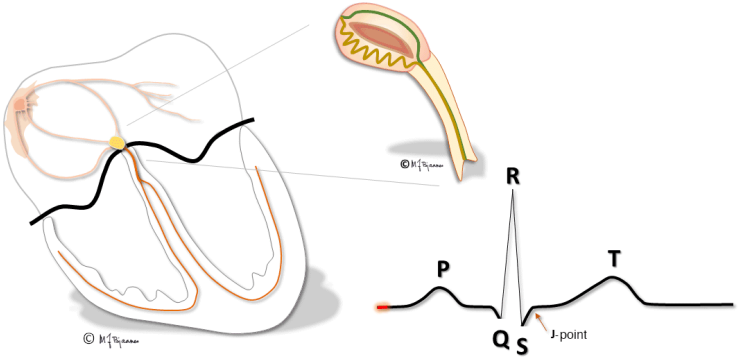
Normal ECG – Lead II

- PR interval
  - Onset of P wave to the start of QRS complex
- PQ Segment
  - End of P wave and start of QRS complex
- ST segment
  - End of S wave and beginning of T wave
    - Interval between ventricular depolarization and repolarization
- QT interval
  - Start of Q wave to end of T wave
    - Time taken for ventricular depolarization and repolarization
      - Period of ventricular contraction to relaxation









Atrial diastole	Atrial systole	Atrial diastole	
Ventricular diastole		Ventricular systole	Ventricular diastole
	One cardiac cycle		

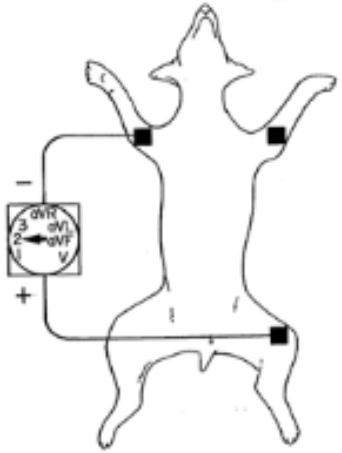
Indications for the electrocardiogram

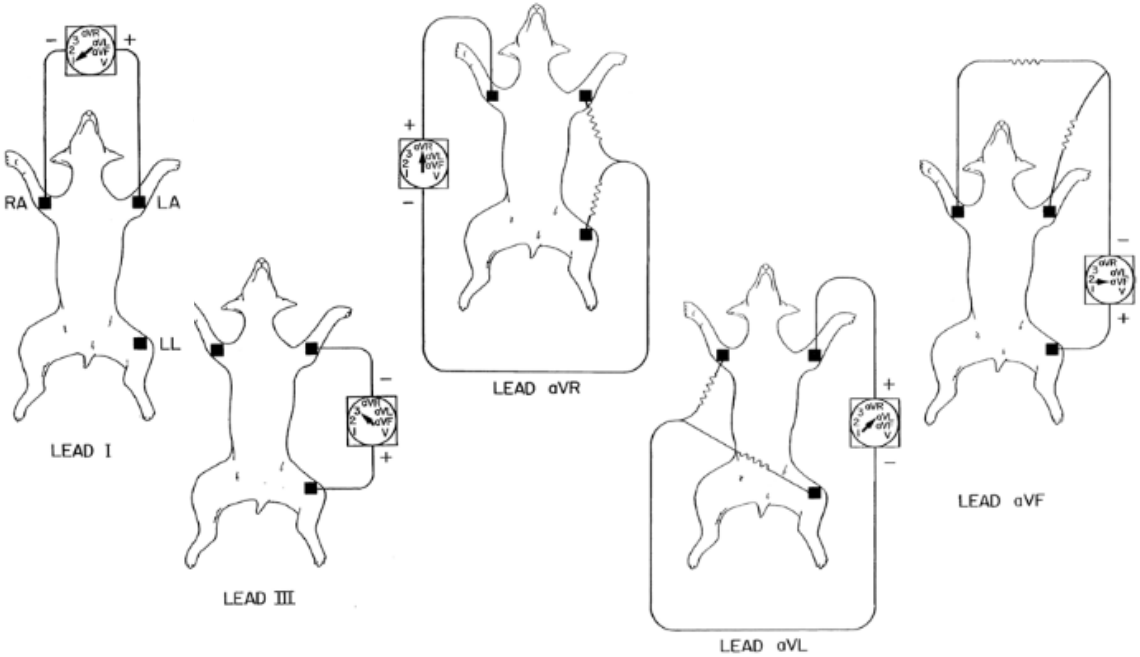
- Arrhythmia
- History of syncope or episodic weakness
- Cardiac chamber enlargement
- Pericardial effusion
- Pericardiocentesis
- Monitoring drug therapy
- Monitoring during anesthesia

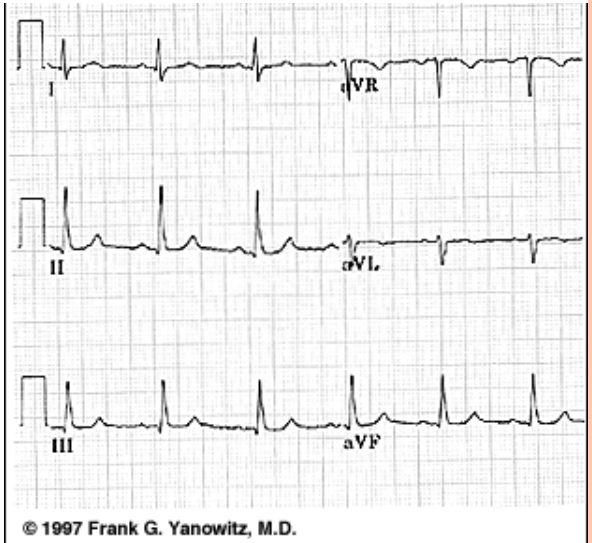
TRACING LEADS

I	right arm to the left arm
II	right arm to the left leg
III	left arm to the left leg
aVR	right arm to a point midway between the left arm and left leg
aVL	left arm to a point midway between the right arm and left leg
aVF	left leg to a point midway between the left arm and right arm

We most commonly use LEAD II

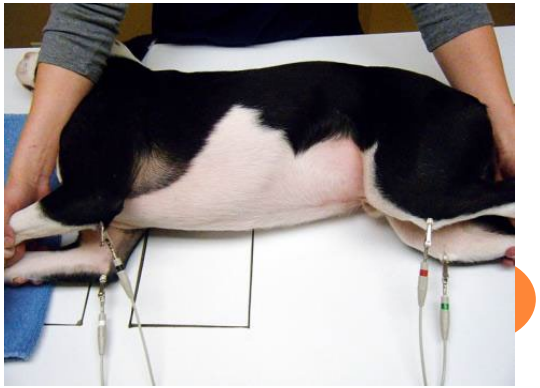






Acquisition of the Electrocardiogram (Limb Leads) ***Right Lateral Recumbency***

- **White electrode—right forelimb (RA)**
- **Black electrode—left forelimb (LA)**
- **Green electrode—right hindlimb (RL)**
- **Red electrode—left hindlimb (LL)**



ELECTROCARDIOGRAM

- **Theory**
  - Measures the electrical depolarization of the heart
- **What It Tells You**
  - Heart Rate
  - Heart Rhythm
  - **Impulse Conduction**
- **What you can infer**
  - Chamber enlargement
  - Myocardial disease
  - Ischemia– lack of O2 causing damage to the muscle tissue
  - Electrolyte imbalances
  - Pericardial disease– fluid in the sac surrounding the heart, thickening of the sac
- **What it Does NOT Tell You**
  - Strength or presence of contractions
  - Cardiac output– how much blood is being pumped
  - Function of chambers or valves
  - Size and Shape of heart
  - **If your heart is beating normally**

Performing an ECG

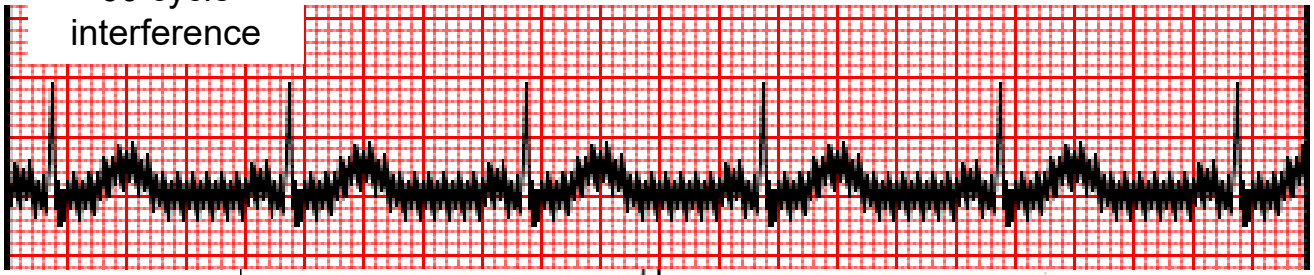
- Keys to success
  - Padding or mat if steel table
  - Conducting gel, or alcohol (avoid if defibrillation is a possibility)
  - Alligator clip or pad (must clip)
  - Right **lateral recumbency** is ideal. (large animals stand, cats can be sternal, if dyspneic—sternal is ok)
  - Challenges: panting, noisy room, other electronics

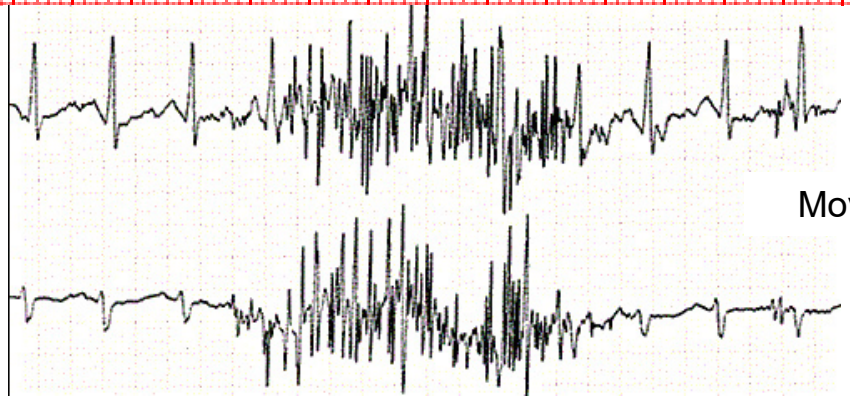
What can complicate an ECG tracing?

- Movement!
  - Shivering patient
  - Respirations (panting = movement)
- Other electrical activity/devices near patient (unplug!)
- Not enough gel or alcohol on alligator clips
- Limbs not perpendicular
- Electrodes in wrong location (should be just distal to elbows and stifles!)
- Patient in wrong position

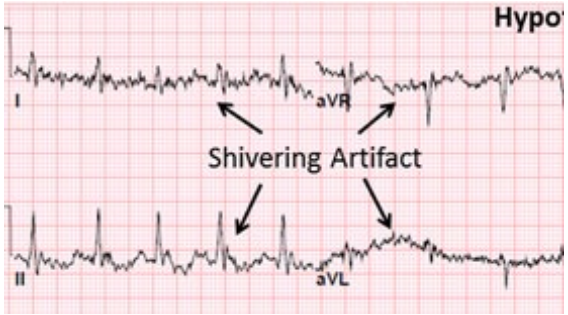
ECG Complications

60 cycle interference



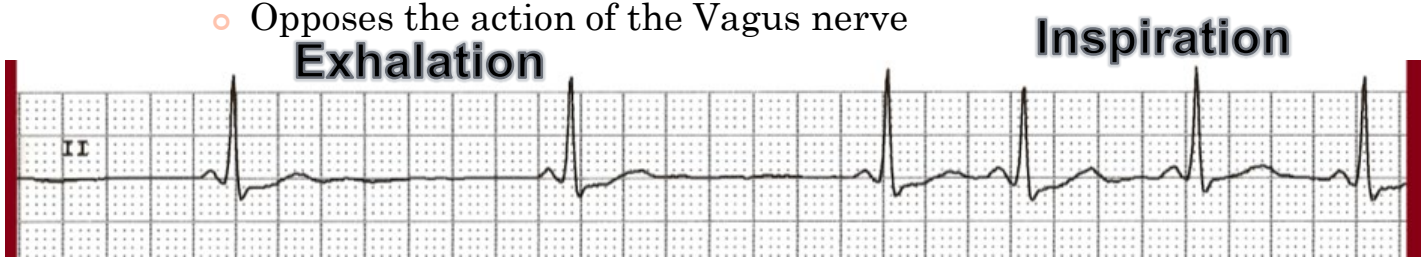


Movement



Heart Rhythm - Normal

- Normal Sinus Arrhythmia (NSA)
  - Rate changes with breathing
  - Caused by ↑ Vagal Tone
    - Stimulates the parasympathetic nervous system ("rest and digest")
    - Lowers heart rate
    - Senses thoracic pressure
      - Negative pressure on inhalation= increased heart rate
      - Positive pressure on exhalation= decreased heart rate
  - Common in Larger or Brachycephalic dogs
  - Can be eliminated with atropine
    - Lowers the parasympathetic nervous system activity
    - Increases firing of the SA node and conduction through the AV node
    - Opposes the action of the Vagus nerve



High Vagal Tone

- Vagus nerve = Cranial Nerve X – controls parasympathetic nervous system to the heart, lungs, GI tract.
- "High vagal tone" = Vagal tone is an internal biological process that represents the activity of the vagus nerve. Increasing your vagal tone activates the parasympathetic nervous system – this usually lowers heart rate and blood pressure.

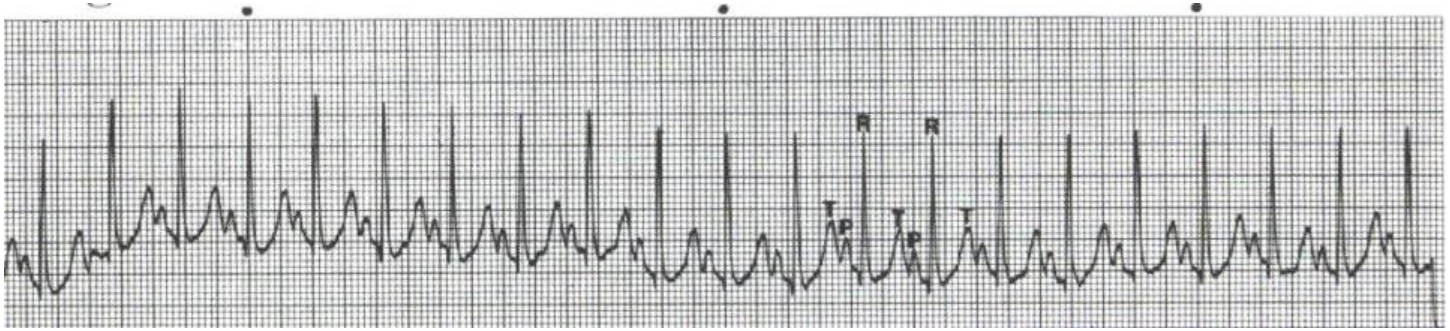
Abnormal Heart Arrhythmias

- Sinus bradycardia
- Sinus tachycardia
- Impulse conduction problems
  - 1st degree, 2nd degree, 3rd degree, Bundle Branch Blocks
- Atrial fibrillation
- Atrial ectopic heartbeats
- Ventricular premature complexes (VPC's)
- Ventricular fibrillation
- Asystole

Heart Rhythm - Abnormal

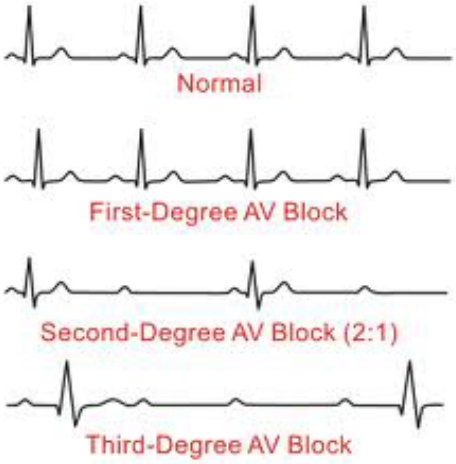
- **Sinus Bradycardia**
  - Dog- <60-70 bpm, Cat- <120 bpm

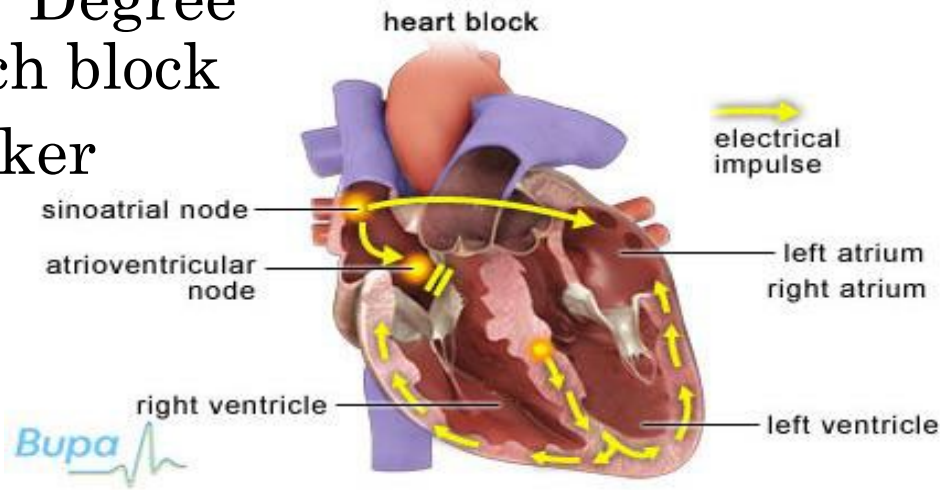
- **Sinus Tachycardia**
  - Dog- >160-180 bpm, Cat- >240



Impulse Conduction Problems

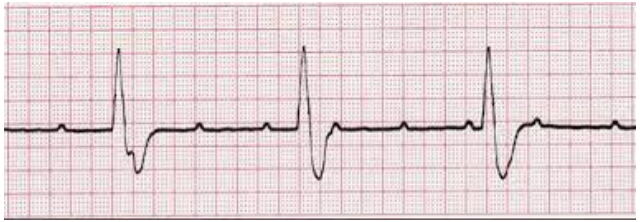
- Where the electrical impulse is blocked from proceeding in the normal fashion through the heart
- Examples: 1st Degree AV block, 2nd Degree AV block, 3rd Degree AV block, Bundle Branch block
- Often require a pacemaker

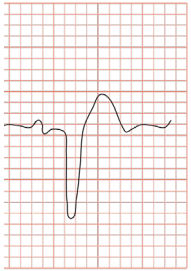




AV Blocks

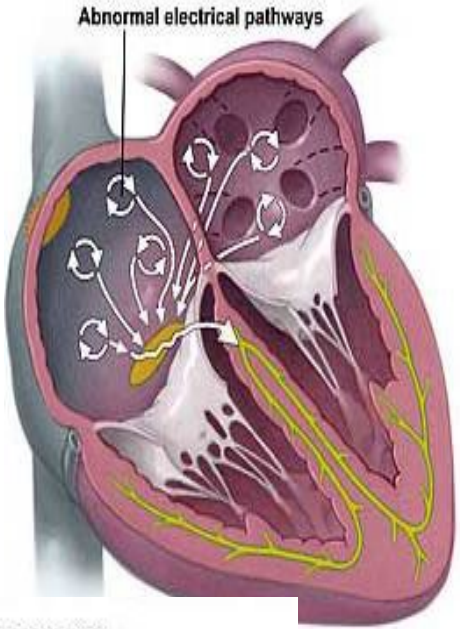
- 1st degree
  - Conduction through the AV node is delayed
  - Prolonged PR interval
- 2nd degree
  - Intermittent disruption of AV node conduction
- 3rd degree
  - No conduction through AV node
- Bundle branch block
  - Blocked at the level of Left or Right bundle branches
  - Slower to depolarize one ventricle
  - QRS complexes are wider than normal



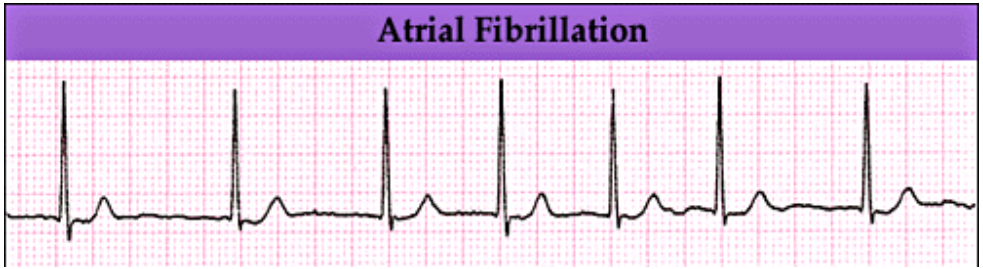


Heart Rhythm - Abnormal

- **Supraventricular, ectopic heartbeat**
  - Depolarization above the ventricles
    - Atrial fibrillation, Atrial premature beats
  - Atria stimulated to contract very quickly and differently from normal activity originating from the SA node
  - Ineffective and uncoordinated contraction of the atria
  - Causes irregular ventricle beats
  - P waves not associated with QRS complex



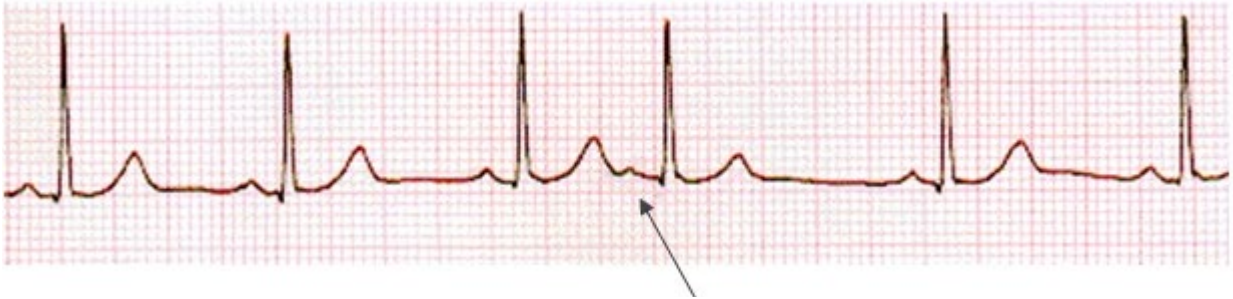
Supraventricular (Ectopic) Arrhythmias



Quivering or irregular atrial beat

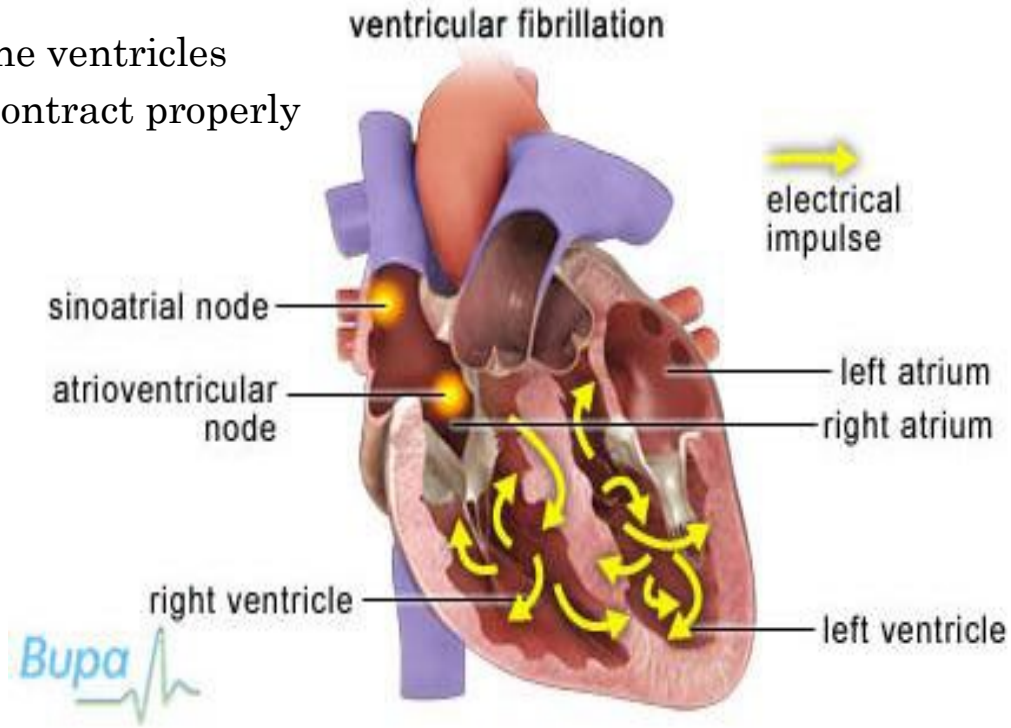
Atrial Premature Contraction

- **Originate in Atria but not SA node**

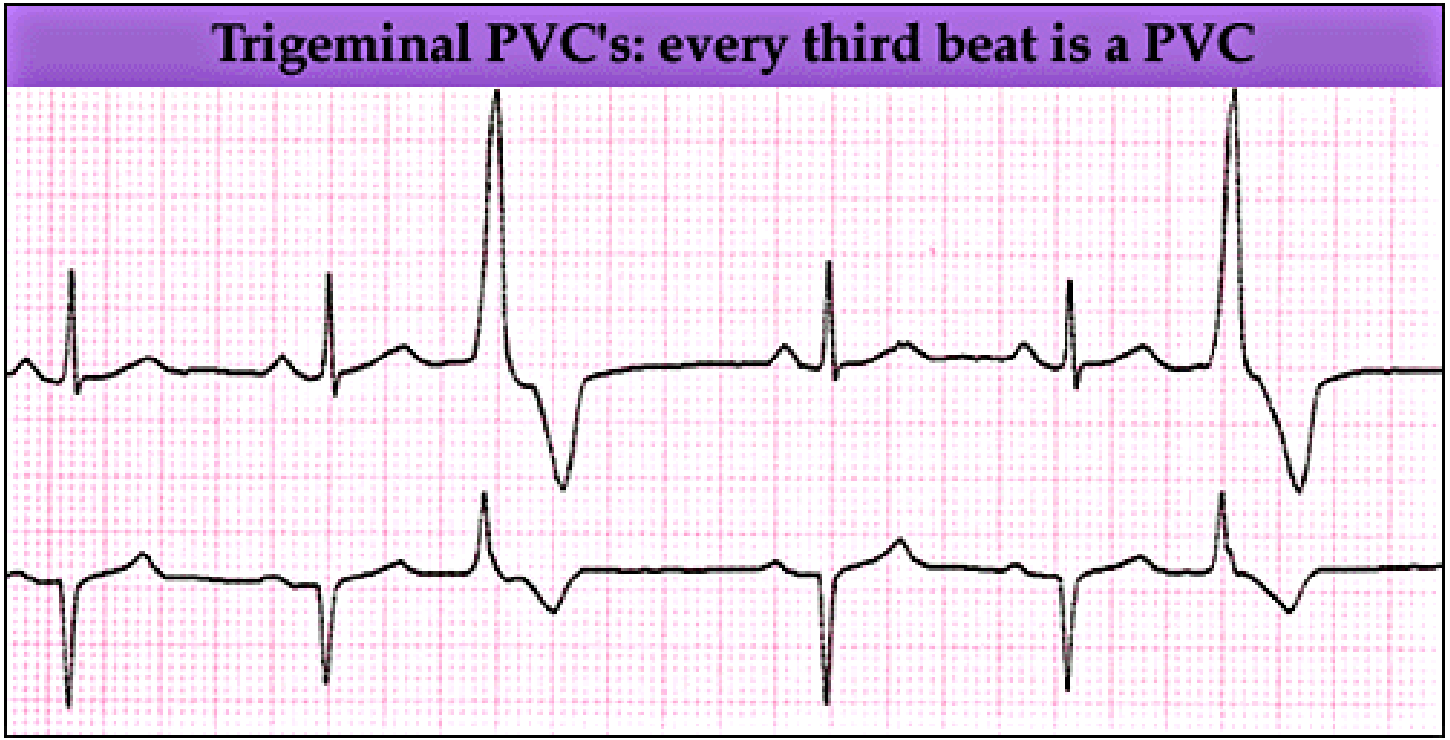


Ventricular Arrhythmias

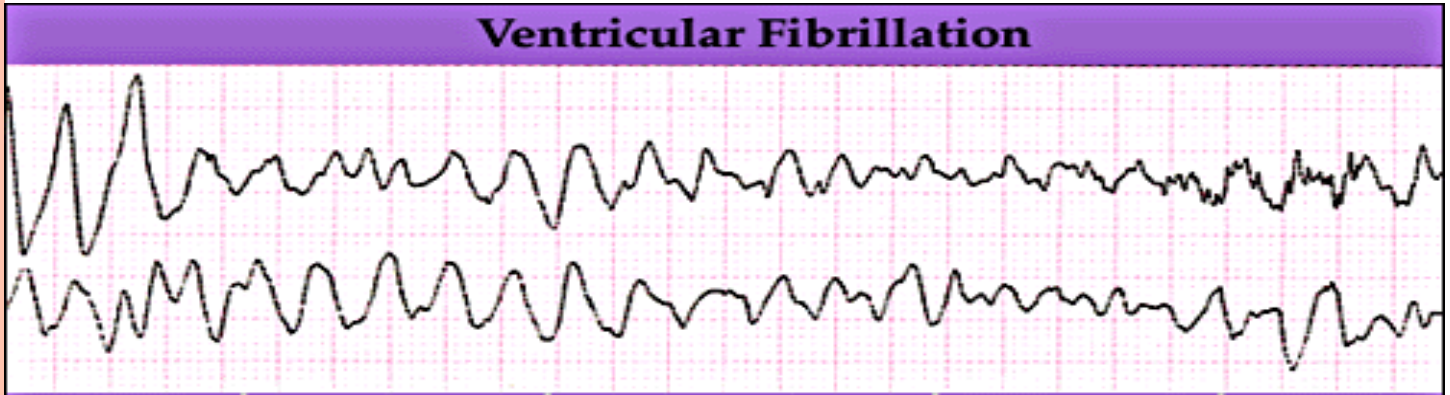
- Abnormal depolarization of the ventricles
  - Ventricular premature complexes
    - Extra heart beats that originate from the ventricles
  - Ventricular fibrillation
    - Uncoordinated contraction of the ventricles
    - Ventricles quiver rather than contract properly
    - Leads to Ventricular asystole



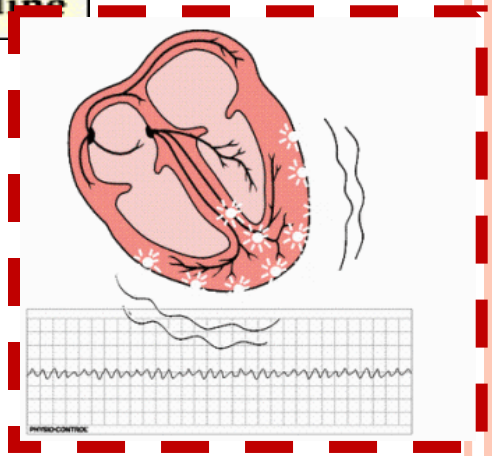
Premature Ventricular Contractions (PVC's or VPC's)



Ventricular Fibrillation/ Asystole



Heart Rate	Rhythm	P Wave	PR interval (in seconds)	QRS (in seconds)
300-600	Extremely irregular	Absent	N/A	Fibrillatory baseline



To sum up Premature Complexes and Fibrillation

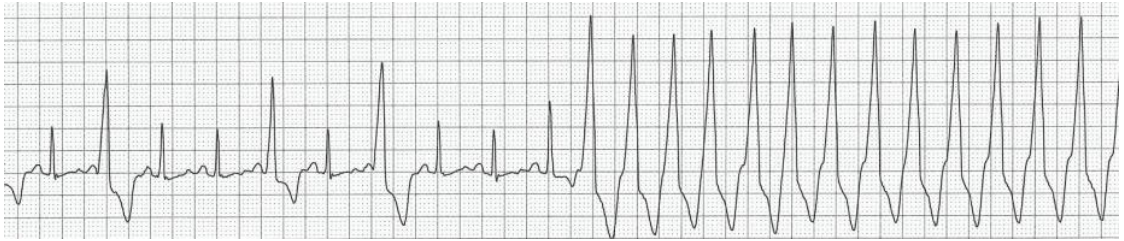
Atrial

- APC's
  - Beats originate in atria but NOT SA node
  - Out of sync with normal sinus rhythm
  - QRS complexes normal
- Atrial tachycardia is 4 or more APC's in a row
- Atrial Fibrillation
  - Irregularly irregular
  - Supraventricular QRS complex
  - No P waves
  - Fibrillation "F" waves present

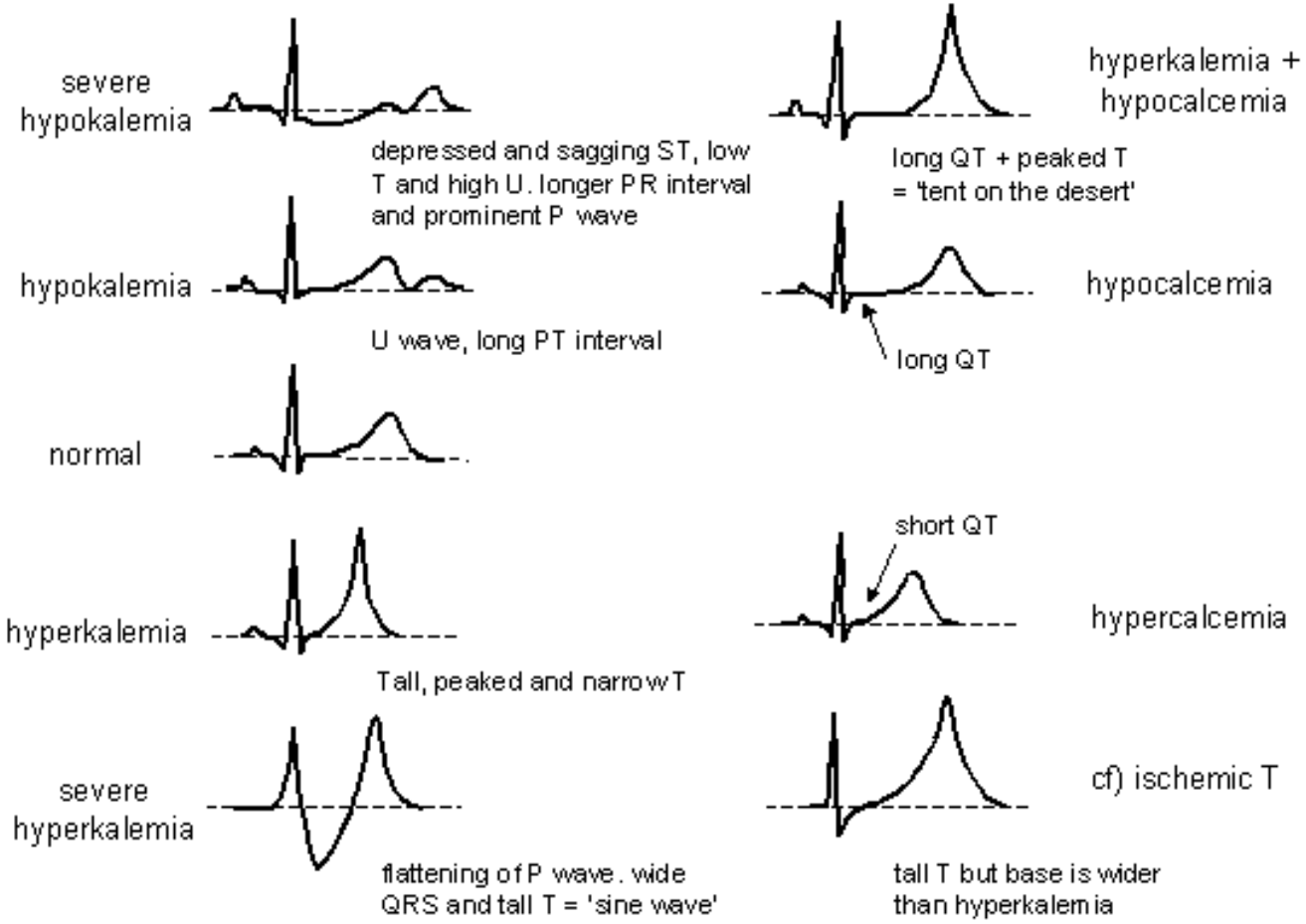
Ventricular

- VPC's
  - Beats originate in ventricle before the SA node can fire
  - Out of sync with normal sinus rhythm
  - Wide and bizarre QRS complexes
  - No P wave
- Ventricular tachycardia is 4 or more VPC's in a row with tachycardia
  - Doctor dependent
- Ventricular fibrillation
  - No organized ventricular activity
  - No QRS complexes

Ventricular spiral of doom



Electrolyte Disturbances



Hyperkalemia

K = 6.0-7.0 mEq/L

"tented" T wave

K = 7.0-8.0 mEq/L

QRS widening

K = 8.0-9.0 mEq/L

PR interval lengthening

K = 9.0-10.0 mEq/L

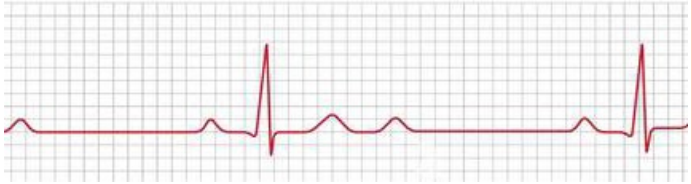
atrial arrest

K>10 mEq/L

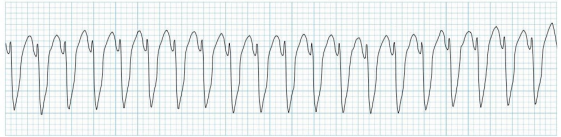
R.I.P.

ECG Interpretation Practice

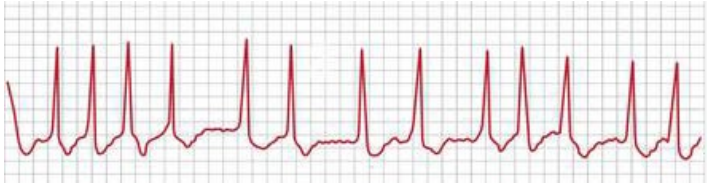
Symmetry is key!!



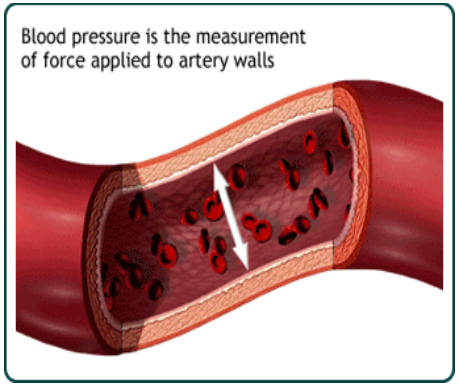
- Are all waves/ complexes present?
  - Do we have P for QRS or QRS for P



- Do they all look the same?
- Is there appropriate spacing of segments?
- Is there appropriate spacing between cycles?
- Great we have an issue is it atrial or ventricular?



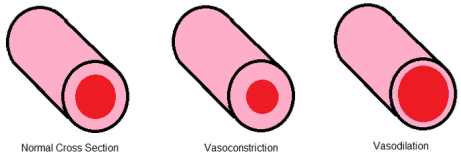
Blood Pressure



- Systole—contraction
- Diastole—relaxation
- Asystole—no contraction (flat line)
- Blood flows from areas of high pressure to low pressure
  - When the ventricles contract, blood flows from high-pressure ventricle to the lower pressure arteries
  - Blood pressure is higher during ventricular systole
  - Higher on the arterial side of the circulatory system than the venous side
  - Lowest pressure is at vena cava's entrance to the right atrium (central venous pressure or CVP)
  - Measured in mmHg

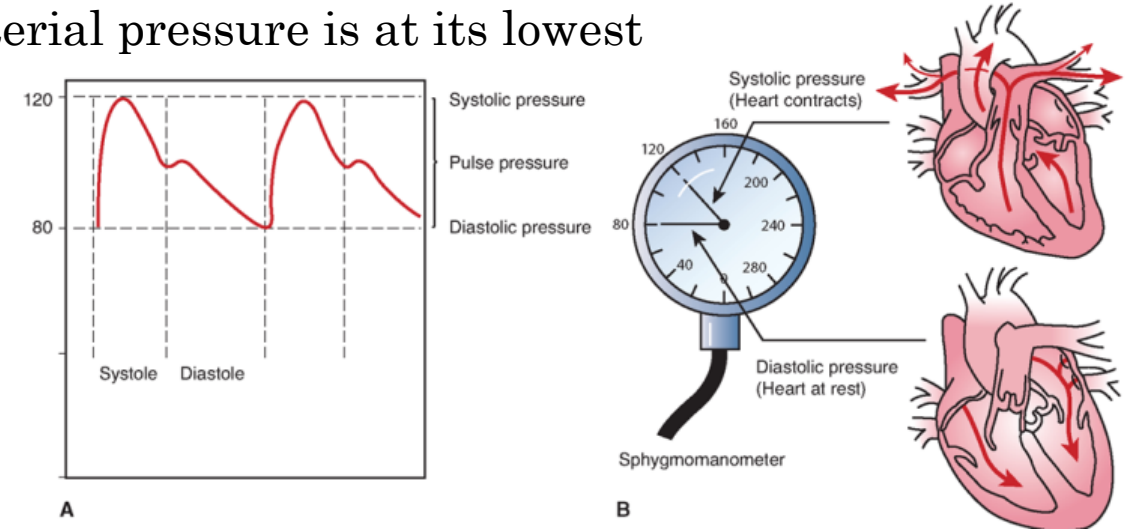
What Effects Blood Pressure

- Resistance
  - A force that opposes the blood flow
  - Smaller vessel = larger resistance
    - Therefore, **vasoconstriction** results in greater resistance, impeding blood flow
    - And **vasodilation** will allow blood to flow more freely (to a point)
- Cardiac output
  - HR, contractility, viscosity of blood



Systolic and Diastolic Pressure

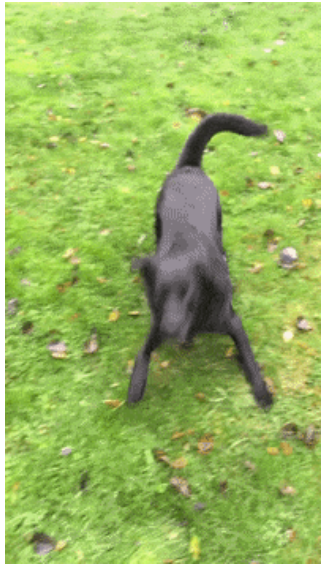
- **Systolic pressure**
  - Pressure of the blood in the arteries during ventricular systole
  - This is when arterial pressure is at its highest
- **Diastolic pressure**
  - Pressure in the arteries during ventricular diastole
  - Arterial pressure is at its lowest



Mean Arterial Pressure

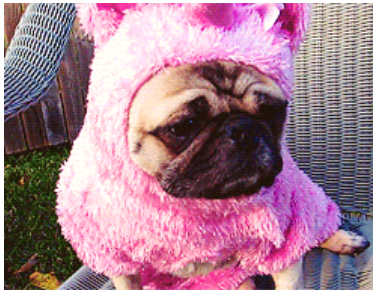
- **Mean Arterial Pressure (MAP)**
  - Average pressure of the blood in the arteries over the entire cardiac cycle. This is calculated or measured.
  - Calculation:
    - MAP= DP + (1/3) (SP- DP)
  - Considered to be the perfusion pressure to the organs of the body
  - 60 mmHg is the lowest pressure able to sustain the organs
    - If the MAP falls significantly below this for too long, the organs will not get enough blood flow and will become ischemic

Normal Blood Pressure Values



Species	Systolic	Diastolic	MAP
Feline	120-140mmHg	50-80mmHg	60-100mmHg
Canine	130-160mmHg	50-80mmHg	60-100mmHg

- Patient's temperament and health needs to be taken into consideration



What affects Blood Pressure?

- Cardiac output
- Blood volume
- Peripheral resistance
  - Increases with vasoconstriction
  - Decreases with vasodilation
- Blood viscosity
  - Viscous blood does not flow freely
  - Viscosity can change due to changes in the concentration of blood cells or plasma proteins
    - Hydration
- Increases in any of these factors will increase blood pressure
- Decreases will decrease blood pressure

Homeostasis

- Blood pressure is maintained within normal limits as part of the body's **homeostatic mechanisms**
  - Pressure receptors respond to changes in arterial pressure
  - They send a signal to the cardiac and vasomotor centers of the brain
  - These centers send impulses via the autonomic nervous system to the heart and vessels
  - Changes occur in the cardiac output or blood vessel diameter in order to bring pressure back towards normal

Hormonal Effects on Blood Pressure

- **Epinephrine and Norepinephrine**
  - **Adrenal gland**
  - Mimic the sympathetic branch of the autonomic nervous system (flight or fight response)
  - Increase blood pressure

- **Antidiuretic hormone (ADH)**
  - **Hypothalamus → Pituitary → Kidney**
  - Reduces fluid loss by retaining H_2O
  - Increases blood volume
  - Increases blood pressure

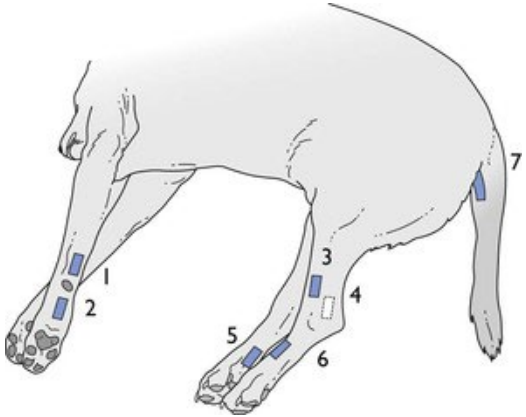
- **Angiotensin**
  - **Liver → Adrenal gland**
  - Causes vasoconstriction, which increases blood pressure
  - Stimulates secretion of the hormone aldosterone
    - Helps conserve sodium, increasing blood volume and blood pressure

Indirect blood pressure monitoring

- Noninvasive
- Most common form of BP measurement
- Doppler
- Oscillometric

Oscillometric

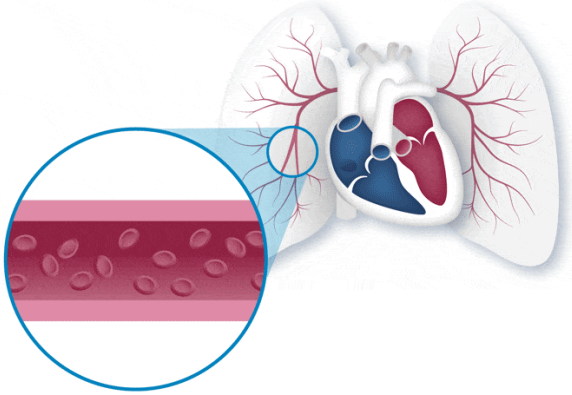
- Systolic, diastolic and MAP
- Automated, less technical skill
- Less accurate
  - Patients under 5 kg, arrhythmias, tachycardia or bradycardia, vasoconstriction, hypothermia, moving or shivering
- Choose appropriate cuff size
- Place proximal to artery

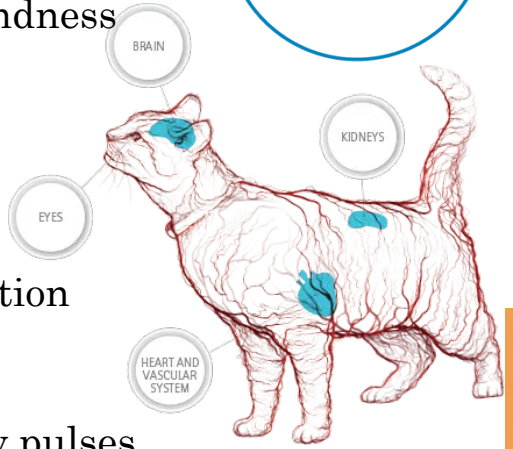


Doppler

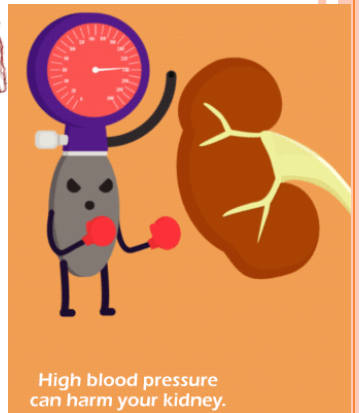
- Systolic only
- Affordable and readily available
- Suitable for patients with hypotension and arrhythmias
- Appropriate BP cuff proximal to artery
  - Width 30-50% of circumference at site of placement
- Clip hair over appropriate artery
  - Dorsal metatarsal, ventral digital, tail
- Conductive gel applied to crystal
- Position directly over artery
  - Listen for clear loud pulse
- Sphygmomanometer attached to cuff
  - Inflate cuff, slowly deflate until pule becomes audible
- Take multiple measurements. Note location and size of cuff in medical record.

- Hypertension
  - Cause
    - Stress, fear, overhydration, metabolic disease (renal, endo, etc.)
  - Signs
    - Increased CRT, injected or brick red mm, bounding pulses, gradual or sudden blindness



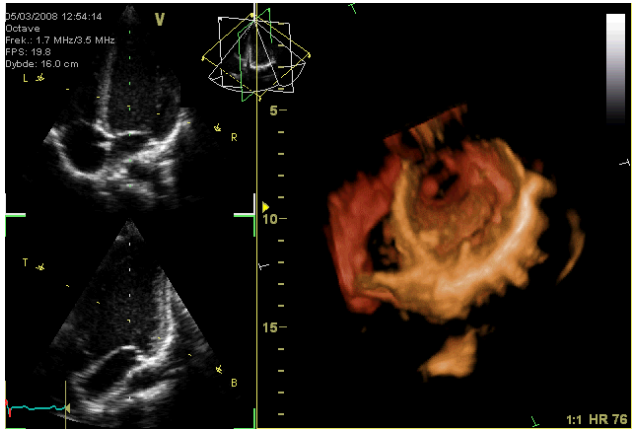


- Hypotension
  - Cause
    - Hypovolemia, vasodilation, myocardial depression, arrhythmias, vagal stimulation (parasympathetic)
  - Signs
    - Pale mm, elongated CRT, weak-thready pulses, decreased HR (sometimes increased), lethargy, collapse, PU/PD, cold extremities, hypothermia



Pulse oximetry (SpO_2)

- Measures the amount of oxygen saturation of red blood cells.
  - Emits and reabsorbs red and infrared light
    - Protect from light/ it may interfere with result
  - Normal values above 95%
    - Below 90%=hypoxia
  - Locations
    - Tongue, lip, prepuce, vulva, digit, etc.
      - Hypotension could affect results
    - Positional, +/- moisture
  - Anesthesia, emergency


05/03/2008 12:54:14
Octave
Frek.: 1.7 MHz/3.5 MHz
FPS: 19.8
Dybde: 16.0 cm
V
L
R
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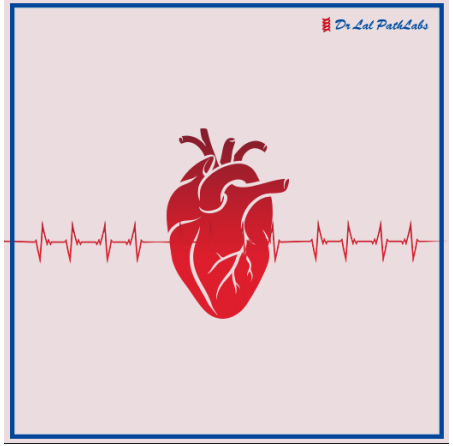
B
5
10
15
1:1 HR 76


Dr Lal PathLabs

THE END